USER & OPERATING MANUAL: DRIER SIZING TENSION CONTROLLER

China FX3U-14MT-6AD-2DA PLC & Fuji Electric FRENIC Inverter Integration

1. System Overview

This control system synchronizes and regulates the drying cylinder (*drier*) speed on a **Textile Warp Sizing Machine**.

The system operates **100% automatically** using a closed-loop **Proportional-Integral (PI)** algorithm: 1. It reads continuous yarn tension from an analog dancer roll (`AD0` : `0 – 10V DC` → `0 – 4000 kg`). 2. It compares the tension against the active tension setpoint (`X0` =1000kg, `X5` =1500kg, `X6` =2000kg, `X7` =2500kg). 3. It filters the tension through an **easily adjustable deadband register (`D125`)** to prevent mechanical hunting. 4. It dynamically calculates the required speed ratio correction (**Parameter `d63`**). 5. It transmits the updated `d63` value directly to a **Fuji Electric Inverter** via **RS-485 Modbus RTU**.

```
[ Dancer Sensor (0-10V) ] ──► [ AD0 Terminal ]
                                      │
[ Selectors: Mode / Setpoint ] ──► [ X0, X2-X7 Terminals ]
                                      │
                                      ▼
                        [ China FX3U-14MT PLC Board ]
                    Calculates d63 = Base_Preset - PI_Trim
                    Deadband easily adjusted via register D125
                                      │
                                RS-485 Modbus
                                (Port 1: A+ / B-)
                                      │
                                      ▼
                          [ Fuji FRENIC Inverter ]
                              (Parameter d63)
                                      │
                                      ▼
                            [ Drier Cylinder Motor ]
```

2. Hardware Specifications

Item	Specification
PLC Model	China FX3U-14MT-6AD-2DA (STM32F103 ARM Cortex-M3 @ 72 MHz)
Power Supply	**DC 24V** ($\pm 10\%$), 1 Amp regulated power supply
Digital Inputs	8 Channels (`X0` – `X7`), NPN Low-Level (Active LOW / Sinking)
Digital Outputs	6 Channels (`Y0` – `Y5`), Transistor NPN Open Collector (DC 24V, 500mA max)

Item	Specification
Analog Inputs	6 Channels (`AD0` – `AD5`), 12-bit resolution (`0 to 4095` counts)
Analog Used	`AD0` : Voltage `0 – 10V DC` connected to Dancer (scaled to `0 – 4000 kg`)
Communication	**COM1:** RS232 Programming/HMI **COM2:** RS-485 Terminals A/B (Modbus RTU Master)
Target Drive	Fuji Electric FRENIC Series (MEGA / Ace / Multi / Mini)

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3. Terminal Wiring Diagram

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                                TOP TERMINAL STRIP
═══════════════════════════════════════════════════════════════════════════════
[ 24V ] ──── Connect to External +24V DC Power Supply (+)
[  0V ] ──── Connect to External  0V DC Power Supply (-)

--- DIGITAL INPUTS (NPN: Connect terminal to 0V / GND to activate) ---
[ X0  ] ──── Setpoint Switch: SETPOINT 1000 kg (Dedicated Target 1000 kg)
[ X1  ] ──── Spare (Unused)
[ X2  ] ──── Machine Mode:    SIPUT FLAG       (Selects Base d63 = 2375 via D205)
[ X3  ] ──── Machine Mode:    SPEED 1 FLAG     (Selects Base d63 = 2300 via D210)
[ X4  ] ──── Machine Mode:    SPEED 2 FLAG     (Selects Base d63 = 2225 via D215)
[ X5  ] ──── Setpoint Switch: SETPOINT 1500 kg (Dedicated Target 1500 kg)
[ X6  ] ──── Setpoint Switch: SETPOINT 2000 kg (Dedicated Target 2000 kg)
[ X7  ] ──── Setpoint Switch: SETPOINT 2500 kg (Dedicated Target 2500 kg)

--- ANALOG INPUTS ---
[ AD0 ] ──── Dancer Analog Signal (+) (0 to 10V DC)
[ AD1 ] ──── Spare (0-10V)
[ AD2 ] ──── Spare (0-10V)
[ AD3 ] ──── Spare (0-20mA)
[ AD4 ] ──── Spare (0-20mA)
[ AD5 ] ──── Spare (0-20mA)
[AGND ] ──── Dancer Analog Signal Ground Reference (-)

═══════════════════════════════════════════════════════════════════════════════
                               BOTTOM TERMINAL STRIP
═══════════════════════════════════════════════════════════════════════════════
--- TRANSISTOR OUTPUTS (DC Loads Only) ---
[ Y0  ] ──── Spare
[ Y1  ] ──── Spare
[ Y2  ] ──── Spare
[ Y3  ] ──── Spare
[ Y4  ] ──── Spare
[ Y5  ] ──── Spare
[COM0 ] ──── Connect to DC 0V (GND)

--- ANALOG OUTPUTS ---
[ DA0 ] ──── Spare Analog Output 1 (0-10V)
[ DA1 ] ──── Spare Analog Output 2 (0-10V)
[AGND ] ──── Analog Ground

--- RS-485 SERIAL PORT ---
[ A+  ] (or D+) ──── Connect to Fuji Inverter RS-485 Terminal (+) / Pin 4 (RJ45)
[ B-  ] (or D-) ──── Connect to Fuji Inverter RS-485 Terminal (-) / Pin 5 (RJ45)
═══════════════════════════════════════════════════════════════════════════════
```

> *[!CAUTION]* ***Analog Ground vs. Power Ground:*** *Always connect the dancer sensor's ground wire to* `AGND`*, never to the digital* `0V` *terminal. This prevents electrical switching noise from corrupting the tension signal.*

4. Fuji Inverter Communication Configuration

Ensure the following parameters are configured on the Fuji Inverter keypad:

Parameter	Setting Name	Value	Description
`y01` / `y11`	Station Address	`1`	Modbus Slave Station ID = 1
`y02` / `y12`	Comms Error Action	`0`	Trip or coast to stop on communication loss
`y03` / `y13`	Comms Error Timer	`1.0`	1.0 second communication error timeout
`y04` / `y14`	Baud Rate	`2`	**9600 bps** (matches PLC register `D8120`)
`y05` / `y15`	Data Length	`0`	**8 Data bits**
`y06` / `y16`	Parity	`1`	**Even Parity**
`y07` / `y17`	Stop Bits	`0`	**1 Stop bit**
`y10` / `y20`	Protocol Selection	`0`	**Modbus RTU Protocol**
`d62`	Pulse Scaling 1	*(Fixed)*	Pulse numerator (set per machine gearing)
`d63`	Pulse Scaling 2	*(Auto)*	**Controlled dynamically by PLC via Modbus**

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5. Control Philosophy & Operating Principles

A. The Sizing Machine Mechanical Dynamics

$$\text{Inverter Frequency } (f^*) = \text{Master Pulses } (Np) \times \left(\frac{\mathbf{d62}}{\mathbf{d63}} \right)$$

Because the dancer roll is positioned **AFTER the drier cylinders** (between Drier and Beamer): * The **Drier FEEDS yarn** into the dancer. * The **Beamer PULLS yarn** out of the dancer.

$$\text{Drier Speed } \propto \frac{1}{\mathbf{d63}}$$

- **When Tension is TOO HIGH ($> \text{Setpoint} + \mathbf{D125}$):**
- The beamer is pulling faster than the drier feeds. Yarn is being stretched.
- The drier must **FEED MORE YARN (SPEED UP)**.
- To speed up the drier, the PLC **DECREASES** `d63` via positive PI Trim.
- **When Tension is TOO LOW ($< \text{Setpoint} - \mathbf{D125}$):**
- The drier is feeding faster than the beamer pulls. Yarn is sagging.
- The drier must **FEED LESS YARN (SLOW DOWN)**.
- To slow down the drier, the PLC **INCREASES** `d63` via negative PI Trim.
- **Within Deadband ($\text{Setpoint} \pm \mathbf{D125}$):**

- Tension is inside the target zone. The PI controller holds steady. **No adjustments are made.**

B. Forward vs. Reverse Action Direction Selector (`M10`)

The Boolean variable `M10` defines the control action direction: * `M10 = 0` **(OFF - Reverse Acting, Default)**:

$$\text{Final } d63 = \text{Base } (D220) - \text{Trim } (D150)$$

Use Case: Standard setup with the dancer located **AFTER** the drier cylinder. High tension ($PV > SV$) decreases $d63$, accelerating the drier to feed more yarn and relieve tension. * `M10 = 1` **(ON - Forward Acting)**:

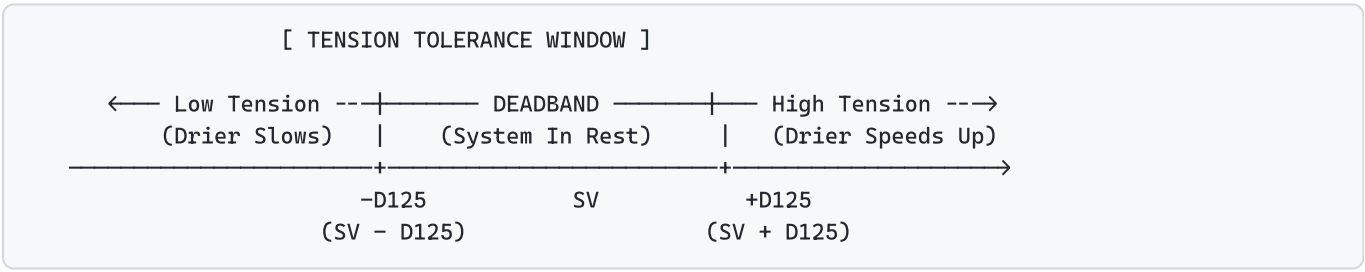
$$\text{Final } d63 = \text{Base } (D220) + \text{Trim } (D150)$$

Use Case: Setup with the dancer located **BEFORE** the drier cylinder (or inverted tension transducer). High tension ($PV > SV$) increases $d63$, decelerating the drier to relieve tension.

Control: `M10` can be toggled via HMI, forced in GX Developer Device Test, or mapped to a physical panel switch (e.g. spare digital input `X001`).

C. Single Master Deadband Architecture (`D125`)

The deadband is controlled by **one single master parameter: Register** `D125` (Default = `100 kg`):



1. **Only ONE Register to Adjust (`D125`):** * The operator or engineer only ever sets or modifies `D125` (e.g. `50` , `70` , `100 kg`). * `D126` **is purely internal:** On every single PLC scan, the ladder automatically calculates the negative boundary:

   $$\texttt{D126} = 0 - \texttt{D125}$$

   * You **never need to touch or configure** `D126` manually.

2. **Inside Deadband ($|\text{PV} - \text{SV}| \leq \mathbf{D125}$):** * **Effective Error** $\texttt{D124} = 0$**:** The control error is completely suppressed. * **Proportional Term** $\texttt{D130} = 0$**:** Instantaneous kick is zero. * **Integral Accumulator** `D140` **Freezes:** Holds its existing steady-state value without accumulating or drifting. * **EEPROM Protection & Motor Stability:** Since $\texttt{D230} = \texttt{D220} - \texttt{D150}$ does not change, $\texttt{D230} == \texttt{D231}$. **Zero RS-485 Modbus packets are sent**, preventing inverter EEPROM wear fatigue and eliminating motor speed jitter.

3. **Bumpless Exit Transfer (No Sudden Jerks):** When tension drifts outside the deadband, the PLC subtracts the deadband offset so correction starts smoothly from zero rather than jumping: * **High Tension ($PV > SV + D125$):**

   $$\text{Effective Error } (D124) = (PV - SV) - D125$$

* **Low Tension ($PV < SV - D125$):**

$$\text{Effective Error } (D124) = (PV - SV) + D125$$

D. Dedicated Speed Modes, Base Presets & Trim Limits

$$\mathbf{d63} = \text{Active_Base_Preset} - \mathbf{PI_{Trim}}$$

The 3 machine speed modes are dedicated to `X002`, `X003`, and `X004`: 1. `SIPUT` **Mode (`X002`):** Base `d63 = 2375` (Saved in `D205`). Lowest speed for crawl/threading. 2. `SPEED 1` **Mode (`X003`):** Base `d63 = 2300` (Saved in `D210`). Normal production speed. 3. `SPEED 2` **Mode (`X004`):** Base `d63 = 2225` (Saved in `D215`). High-speed production.

The **PI Trim (`D150`)** continuously floats on top of whichever mode is selected:

$$\text{PI_Trim} = \text{P_Term} + \text{I_Accumulator}$$

* **P-Term:** Instantly reacts to tension displacement. * **I-Accumulator:** Gently accumulates every 200 ms to eliminate long-term tension offset.

E. Dedicated Tension Setpoint Selection (`X0`, `X5`, `X6`, `X7`)

Terminal	Target Tension Setpoint	Register `D120` Value
`X000`	**1000 kg**	`D120 = 1000` (Also power-up default)
`X005`	**1500 kg**	`D120 = 1500`
`X006`	**2000 kg**	`D120 = 2000`
`X007`	**2500 kg**	`D120 = 2500`

F. Built-In Safeguards

1. **VFD EEPROM Memory Protection:** The PLC compares current `d63` with the last sent value (`LD<> D230 D231`). Modbus transmission is triggered **only when the value changes**, preventing the inverter's memory chip from wearing out.
2. **Web-Break / Line-Stop Lockout:** If the yarn snaps or the line stops, the dancer drops to 0 kg. The PLC checks if `Tension > 150 kg`. If tension is below 150 kg, the PI controller freezes and resets `Trim = 0` so the drier will **never run away to high speed**.
3. **Integral Anti-Windup Clamping:** Both the integral accumulator (`D140`) and the total PI trim (`D150`) are dynamically clamped against the active speed mode limits (`[D224, D222]`), preventing wind-up during rapid line acceleration or bobbin changes.

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6. Complete Memory Register Reference

Device	Name / Function	Range / Default	Description
`X000`	`SETPOINT 1000KG`	Digital Input	Dedicated 1000 kg tension selector
`X001`	Spare	—	Unused
`X002`	`SIPUT FLAG`	Digital Input	Dedicated Siput mode selector (Base `d63 = 2375`)
`X003`	`SPEED 1 FLAG`	Digital Input	Dedicated Speed 1 mode selector (Base `d63 = 2300`)
`X004`	`SPEED 2 FLAG`	Digital Input	Dedicated Speed 2 mode selector (Base `d63 = 2225`)
`X005`	`SETPOINT 1500KG`	Digital Input	Dedicated 1500 kg tension selector
`X006`	`SETPOINT 2000KG`	Digital Input	Dedicated 2000 kg tension selector
`X007`	`SETPOINT 2500KG`	Digital Input	Dedicated 2500 kg tension selector
`M2`	Siput Flag	Internal Relay	Active when in Siput mode
`M3`	Speed 1 Flag	Internal Relay	Active when in Speed 1 mode
`M4`	Speed 2 Flag	Internal Relay	Active when in Speed 2 mode
`M10`	**PI Direction Selector**	**0=REV, 1=FWD**	**Boolean: 0 = Reverse (After Drier), 1 = Forward (Before Drier)**
`M50`	Modbus Trigger	Internal Relay	Pulses ON when new `d63` needs transmission
`M100`	`SIMULATOR BYPASS RD3A`	Internal Relay	Turn ON in GX Simulator to test D100 freely without RD3A overwriting
`T20`	PI Sampling Timer	200 ms	Controls the integration clock rate
`D100`	Raw ADC Count	0 to 4095	Raw 12-bit analog input from terminal `AD0`
`D104`	Unfiltered Tension	0 to 4000 kg	Raw tension calculated from `AD0` via 32-bit `DDIV`
`D106`	DDIV Remainder	-	32-bit division remainder output
`D108`	Filter Temp Register	-	Intermediate calculation register for filter
`D110`	Filtered Tension (PV)	0 to 4000 kg	Clean 4-sample running average of dancer tension
`D120`	Tension Setpoint (SV)	1000 – 2500 kg	Selected tension target
`D122`	Raw Error	$\mathrm{PV} - \mathrm{SV}$	Deviation from setpoint
`D124`	Effective Error	Error after Deadband	Error fed into PI algorithm (0 inside deadband)
`D125`	**Deadband Setting**	**100 kg**	**Tuning register: easily adjust deadband live!**
`D126`	Negative Deadband	Calculated (`-D125`)	Automatic negative threshold (`0 - D125`)

Device	Name / Function	Range / Default	Description
`D128`	P Product 32-bit	—	Intermediate multiplication register
`D130`	Proportional Term	-80 to $+80$	$(\text{Error} \times K_p)/100$
`D132`	I Product 32-bit	—	Intermediate multiplication register
`D134`	Integral Increment	—	Delta step per 200ms
`D140`	Integral Sum	-80 to $+80$	Accumulated steady-state correction
`D150`	Total Dynamic Trim	-100 to $+100$	Sum of $P + I$
`D160`	**CAL AD0 (0 kg)**	**310**	**Retentive calibration register: 0 kg raw count (0.82V slack)**
`D161`	**CAL AD1 (1000 kg)**	**640**	**Retentive calibration register: 1000 kg count (1.62V, X000)**
`D162`	**CAL AD2 (2000 kg)**	**970**	**Retentive calibration register: 2000 kg count (2.42V, X006)**
`D163`	**CAL AD3 (3000 kg)**	**1220**	**Retentive calibration register: 3000 kg count (3.15V)**
`D164`	**CAL AD4 (4000 kg)**	**1400**	**Retentive calibration register: 4000 kg count (3.57V top limit)**
`D168`	Calibration Span AD	Dynamic	Intermediate span calculation register
`D169`	32-bit High Word (0)	0	Cleared to 0 for DDIV double word division
`D205`	Base Siput `d63`	2375 (`H947`)	Base ratio for Siput mode
`D206`	**Siput Trim Limit**	**60**	**Max trim range for Siput mode (+/- 60)**
`D210`	Base Speed 1 `d63`	2300 (`H8FC`)	Base ratio for Speed 1 mode
`D211`	**Speed 1 Trim Limit**	**100**	**Max trim range for Speed 1 mode (+/- 100)**
`D215`	Base Speed 2 `d63`	2225 (`H8B1`)	Base ratio for Speed 2 mode
`D216`	**Speed 2 Trim Limit**	**120**	**Max trim range for Speed 2 mode (+/- 120)**
`D220`	Active Base `d63`	2000 – 3000	The base ratio for currently active mode
`D222`	Active Positive Trim Limit	Dynamic	Positive clamp limit (+D206, +D211, or +D216)
`D224`	Active Negative Trim Limit	Dynamic	Negative clamp limit (-D206, -D211, or -D216)
`D230`	Final `d63` Output	$\text{Base} - \text{Trim}$	**Value written to Fuji Inverter Parameter** `d63`
`D231`	Last Sent `d63`	2000 – 3000	History register to detect value changes
`D320`	Proportional Gain K_p	**5**	Tuning register for immediate response
`D322`	Integral Gain K_i	**2**	Tuning register for steady-state offset removal
`D8120`	RS-485 Config	`H8087`	9600 bps, 8 Data, Even Parity, 1 Stop bit

Device	Name / Function	Range / Default	Description
`D8129`	Modbus Timeout	500 ms	Serial communication timeout

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7. How to Load the Program into the PLC

Using Mitsubishi GX Works2:

1. Open **GX Works2**.
2. Go to **Project** → **New...** * **Series:** `FXCPU` * **Type:** `FX3U / FX3UC` * **Project Type:** `Simple Project` (Ladder)
3. Connect your PC to the PLC's programming port using a USB-to-RS232 programming cable (COM port typically set to **38.4 kbps**, 7-E-1).
4. Go to **Project** → **Open Other Format Data** → **Import from Text File...** * Select: `export/MAIN export.txt`
5. Go to **Edit** → **Documentation** → **Import from Text File...** * Select: `export/COMMENT export.txt`
6. Press `F4` (Compile / Rebuild All).
7. Go to **Online** → **Write to PLC...**, check **Program** and **Comment**, and click **Execute**.
8. Turn the PLC RUN/STOP switch to **RUN**. The `RUN` LED should glow solid green.

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8. On-Site Tuning Guide

8.1 Procedure to Define the Perfect Base Preset (D205, D210, D215)

In sizing machines, the dancer arm should only perform fine-trimming (5%–10%), while the Base Preset carries the main synchronized speed (90%–95%).

```
              [ Ideal Operating Condition ]
Base Preset (D220)  ═══⟹ Carries 100% of nominal speed
PI Trim     (D150)  ═══⟹ Hovers near 0 (dancer rests in center)
Output      (D230)  ═══⟹ D220 - 0 = D220
```

Step-by-Step Machine Calibration:

1. **Run the Machine** in the selected speed mode (e.g. Speed 1 with `X003 = ON`).
2. **Select Tension Setpoint** (e.g. 1000 kg with `X000 = ON`).
3. **Wait for Yarn to Stabilize** at normal running speed and moisture.
4. **Observe Register `D150` (Total Trim)** in GX Works2 monitor (`F3`): - `D150` **floats around 0 (-10 to +10)**: The Base Preset is **perfect**. The dancer is riding in the middle of its physical stroke. - `D150` **consistently sits at +30 to +60**: The drier is feeding yarn too slowly $\implies$ needs more speed $\implies$ $d63$ must be lowered.
    - *Correction*: Lower Base Register `D210` by the offset:

$$\text{New } D210 = \text{Old } D210 - D150$$

(e.g., $2300 - 40 = \mathbf{2260}$*).*

- `D150` **consistently sits at -30 to -60**: The drier is feeding yarn too quickly $\implies$ needs less speed $\implies$ $d63$ must be raised.
- *Correction*: Increase Base Register `D210` by the offset:

$$\text{New } D210 = \text{Old } D210 + |D150|$$

(e.g., $2300 + 40 = \mathbf{2340}$*).*

5. Once dialed in, `D150` will float near `0`, leaving the full trim authority available for unexpected line shocks.

8.2 Adjusting Mode Trim Limits (D206, D211, D216)

Each speed mode has its own independent trim limit register: - `D206` **(Siput Trim Limit, Default = 60)**: Clamps trim to ± 60 during crawl. Keeps slow threading smooth without jerks. - `D211` **(Speed 1 Trim Limit, Default = 100)**: Clamps trim to ± 100 during standard production. - `D216` **(Speed 2 Trim Limit, Default = 120)**: Clamps trim to ± 120 during high-speed production. - *Open-Loop Option*: Set any mode's trim limit register to `0` to completely disable dancer trim and run strictly on fixed base speed.

You can monitor and adjust the controller directly from GX Works2 Online Monitor (`F3`):

8.3 Live Recalibrating the Tension Sensor (D160–D164)

The 5-segment piecewise linear calibration is fully held in retentive registers `D160` to `D164`. If you ever replace the dancer potentiometer or the sensor mechanical alignment shifts, **you do not need to modify any ladder logic**: 1. Leave dancer completely slack (0 kg) -> read raw count in `D100` -> write into `D160` (Default = 310, 0.82V). 2. Hang or pull 1000 kg tension (`X000`) -> read `D100` -> write into `D161` (Default = 640, 1.62V). 3. Hang or pull 2000 kg tension (`X006`) -> read `D100` -> write into `D162` (Default = 970, 2.42V). 4. Pull 3000 kg tension -> read `D100` -> write into `D163` (Default = 1220, 3.15V). 5. Pull 4000 kg tension -> read `D100` -> write into `D164` (Default = 1400, 3.57V).

Automatic Midpoint Calibration: - **1500 kg Setpoint (`X005`)**: Hits the exact midpoint of Segment 2 (**805 counts / 2.02V**) with **0.0 kg error**. - **2500 kg Setpoint (`X007`)**: Hits the exact midpoint of Segment 3 (**1095 counts / 2.78V**) with **0.0 kg error**.

The PLC ladder automatically recalculates all span slopes every scan cycle!

8.4 Adjusting the Single Master Deadband (`D125`)

Now that the 5-segment piecewise calibration provides high-precision tension measurement (0.0 kg error at all 4 setpoints), the deadband can be tuned with great confidence using **single register** `D125` :

```
      REGISTER                DEFAULT          RECOMMENDED TUNING RANGE
-------------------------------------------------------------------------
 D125 (Master Deadband)       100 kg           50 kg to 120 kg (+/- tolerance)
 D126 (Negative Boundary)    -100 kg           Auto-calculated (0 - D125)
```

- **Standard Default:** `D125 = 100` (Tolerance: ± 100 kg).

- **Fine Yarn / High Quality Sizing:** Change `D125` to `50` (± 50 kg) or `70` (± 70 kg) for tighter physical tension regulation.
- **Coarse Yarn / Heavy Beams with Natural Flutter:** If mechanical vibrations or yarn ripples cause excessive minor trimming, raise `D125` to `120` or `150` to let the machine run smoothly inside tolerance without reacting to harmless mechanical flutter.
- **Reminder:** You only ever change `D125`. Register `D126` updates automatically every PLC scan cycle.

8.5 Adjusting Controller Gains (`D320` and `D322`)

```
       DANCER RESPONSE BEHAVIOR                        WHAT TO ADJUST
┌──────────────────────────────────────────────────────────────────────┐
│ The dancer reacts too slowly to line speed changes,                  │
│ and tension wanders outside the deadband for too long:               │
│ ──► Increase Kp (D320) from 5 to 7 or 8.                             │
├──────────────────────────────────────────────────────────────────────┤
│ The dancer is too aggressive, starts bouncing or vibrating,          │
│ and causes the drier speed to oscillate:                             │
│ ──► Decrease Kp (D320) from 5 to 3 or 4.                             │
├──────────────────────────────────────────────────────────────────────┤
│ The dancer settles smoothly, but stays offset just outside           │
│ the deadband without returning to center:                            │
│ ──► Increase Ki (D322) from 2 to 3 or 4.                             │
├──────────────────────────────────────────────────────────────────────┤
│ Tension overshoots the setpoint when returning to center:            │
│ ──► Decrease Ki (D322) from 2 to 1.                                  │
└──────────────────────────────────────────────────────────────────────┘
```

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9. Troubleshooting & Diagnostics

Symptom	Probable Cause	Action
`ERR` **LED is flashing on PLC**	Hardware watchdog or syntax fault	Check DC 24V power supply voltage. Ensure memory range settings in GX Works2 match FX3U.
Dancer reads `0 kg` **(** `D100 = 0` **)**	Wiring broken or 0-10V signal missing	Measure DC voltage between terminal `AD0` and `AGND` with a multimeter. Ensure sensor power supply is ON.
Dancer reads `4000 kg` **constantly**	Open ground wire or input saturation	Check `AGND` terminal connection. Ensure input signal does not exceed 10.0V DC.
Inverter does not change speed	RS-485 wiring reversed or baud mismatch	1. Swap terminals `A+` and `B−` on RS-485 port. 2. Verify Inverter parameters `y01 = 1`, `y04 = 2` (9600 bps), `y06 = 1` (Even), `y10 = 0` (Modbus).
Inverter trips on Comms Fault (`ErP` **)**	Timeout threshold too strict	Check inverter parameter `y03` (set to `1.0` second or higher). Ensure cable shielding is grounded at one end.
Tension increases when drier accelerates	Inverted dancer physical placement	If dancer was physically moved *before* the drier instead of after, change `SUB D220 D150 D230` to `ADD D220 D150 D230` in Section 7.